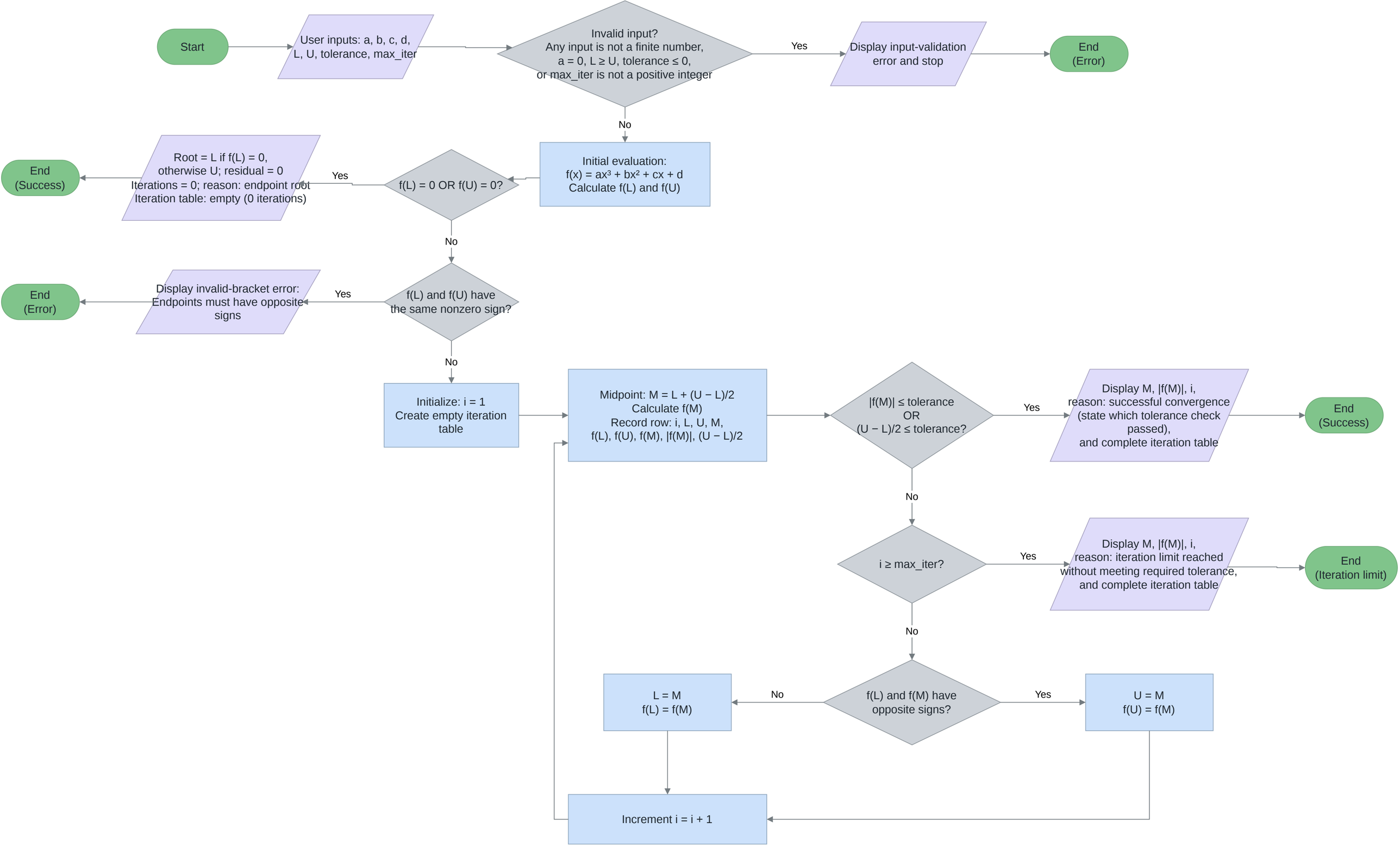
Start
User inputs: a, b, c, d, L, U, tolerance, max_iter
Invalid input? Any input is not a finite number, a = 0, L ≥ U, tolerance ≤ 0, or max_iter is not a positive integer
Yes
Display input-validation error and stop
End (Error)
No
Initial evaluation: f(x) = ax³ + bx² + cx + d Calculate f(L) and f(U)
f(L) = 0 OR f(U) = 0?
Yes
Root = L if f(L) = 0, otherwise U; residual = 0 Iterations = 0; reason: endpoint root Iteration table: empty (0 iterations)
End (Success)
No
f(L) and f(U) have the same nonzero sign?
Yes
Display invalid-bracket error: Endpoints must have opposite signs
End (Error)
No
Initialize: i = 1 Create empty iteration table
Midpoint: M = L + (U − L)/2 Calculate f(M) Record row: i, L, U, M, f(L), f(U), f(M), |f(M)|, (U − L)/2
|f(M)| ≤ tolerance OR (U − L)/2 ≤ tolerance?
Yes
Display M, |f(M)|, i, reason: successful convergence (state which tolerance check passed), and complete iteration table
End (Success)
No
i ≥ max_iter?
Yes
Display M, |f(M)|, i, reason: iteration limit reached without meeting required tolerance, and complete iteration table
End (Iteration limit)
No
f(L) and f(M) have opposite signs?
Yes
U = M f(U) = f(M)
No
L = M f(L) = f(M)
Increment i = i + 1